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Comparing Generative AI and teacher feedback: student perceptions of usefulness and trustworthiness

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ABSTRACT

The rapid integration of Generative Artificial Intelligence (GenAI) into educational contexts has presented both opportunities and challenges for students seeking and using feedback. While AI-generated feedback can offer increased access, timely responses and personalised insights, concerns about the quality of AI-generated feedback still persist, including issues of bias, factual inaccuracies, and homogenisation. This study investigates how students use, value and trust AI-generated feedback compared to feedback from educators. This paper draws on a large-scale cross-sectional survey administered across four major Australian universities. A quantitative analysis of 6960 respondents revealed that half of the students sought feedback from GenAI. Overall, students reported teacher feedback to be more helpful, and especially trustworthy. This paper also reports on the thematic analysis of 8642 open-ended responses, providing deeper insights into students' experiences. This includes findings that students valued the feedback from GenAI because of its ease of access, timeliness, volume, understandability and that it was perceived to be less risky than seeking feedback from teachers. At the same time, students were concerned about GenAI's reliability as well as contextual and disciplinary expertise. We argue that GenAI and teacher feedback appear to serve different needs, and therefore are complementary but not interchangeable.

KEYWORDS

Generative artificial intelligence; feedback; survey; higher education

Introduction

In recent decades, there has been a paradigm shift in feedback scholarship, with a focus on feedback as a student-centred process (Carless 2015). Along with this shift has been the recognition that we need to pay attention to not just the nature of the feedback information, but also how and why students engage with it, including how they make sense of and use this information. The rapid adoption of generative artificial intelligence (GenAI) therefore has significant implications for feedback in higher education. For example, a growing body of research indicates AI-generated feedback has the potential to provide timely and personalised feedback information

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(Chan and Hu 2023; Lee and Moore 2024; Shi and Aryadoust 2024; Nazaretsky et al. 2024). However, it is also well recognised that GenAI's statistical genesis promotes biases and other limitations, including the risk of factual inaccuracies, hallucinations, cultural colonisation, and a tendency towards 'expected' ideas and expressions (Creely and Henderson 2025; Gašević, Siemens & Sadiq, 2023). In a landscape where GenAI is being rapidly adopted, there is an urgent need to better understand how and why students use, value and trust AI-generated feedback compared to feedback from educators. To address this need, this Australian cross-institutional large-scale study employs a survey methodology to describe students' experience of self-generated GenAI feedback, and how this compares with teacher feedback.

Feedback and AI

Feedback is a contested topic in educational research. Broadly speaking, feedback can be seen as: information to a learner about their work or performance; or the interactions surrounding this information; or as the impact that this information has upon the learner. Carless (2015) calls the latter approach 'new paradigm' and this is exemplified by the definition: 'Feedback is a process in which learners make sense of information about their performance and use it to enhance the quality of their work or learning strategies.' (Henderson, Phillips, et al., 2019, p. 1402). This study has been informed by this definition, but we also recognise that our respondents most likely think of feedback as the information they receive from an appraisal of their work. Irrespective of how feedback is understood, feedback can have a powerful influence on student learning and experience, but it is also complex and difficult to implement (Henderson, Ryan, et al., 2019; Leenknicht and Carless 2023).

In part to address these concerns, there is a rising emphasis on how a learner agentically seeks and uses information about their performance. Thus, the concept of 'student feedback literacy' may be particularly relevant to how learners interact with GenAI, as they may frequently do so without any guidance from a teacher. Carless and Boud (2018) conceptualise feedback literacy as having three aspects that directly influence the learner's actions, roughly: understanding feedback processes, making evaluative judgements, and managing 'feelings, emotions and attitudes' (p. 1317). In other words, how the student *feels* matters. Interestingly, a realist review of feedback on written tasks suggests that, while negative emotions can reduce motivation, such effects can be mitigated through building interpersonal trust and relatedness (Ajjawi et al. 2022). This highlights the potential complexity of GenAI feedback in an already complex space.

While studies that explore student perspectives are well established in the broader feedback research, they are largely absent from studies examining GenAI feedback (Nazaretsky et al. 2024). In their systematic review Shi and Aryadoust (2024) note that there is a growing body of research exploring the effectiveness of GenAI feedback, but there is a need for research that explores more complex issues around feedback such as anxiety, motivation, self-regulation and factors that influence perception and use. While less prolific, there are some interesting glimpses of the ways in which student perspectives are being explored, or at least, being identified as significant moderators of GenAI feedback experience. For example, Lee and Moore (2024) systematic review identified several studies that demonstrated GenAI feedback could serve multiple instructional purposes and roles, including dialogue partner, teaching assistant and emotional support. Another example is Ziqi et al. (2024) study which highlighted significant barriers to effective uptake of GenAI feedback by students, including perceptions of vagueness and irrelevance.

There is also a small but interesting number of studies that directly compare GenAI and human feedback. A number of studies have shown that the use of GenAI feedback had a positive impact on student performance (Shi and Aryadoust 2024; Wang 2024), and were 'in some cases even comparable to well-trained human evaluators' (Nazaretsky et al. 2024, p. 285). In addition to supporting improved writing, Wang (2024) found that learners who received GenAI feedback rated lowest on anxiety scales, followed by the teachers' group. However, there are also studies that are not so

positive. In their systematic review of AI-based written feedback, Shi and Aryadoust (2024) stated: 'no conclusive evidence was found to support the effectiveness of [Automated Written Feedback] in comparison to teacher feedback' (p. 202). Interestingly, they note that this does not necessarily suggest that GenAI feedback is not useful, but that it may indicate that GenAI and Teacher feedback may serve different needs and therefore direct comparisons may be misleading.

Nazaretsky et al. (2024) study asked students to evaluate feedback in authentic educational settings both before and after disclosing that the feedback had been created by a teacher or GenAI. They found that disclosing the identity of the feedback provider affected students' perceptions, revealing that students preferred human-created feedback and rated AI-generated feedback as being less credible than feedback provided by their teachers. This echoes studies that show that, when students are given exactly the same feedback will regard it less highly if they think it is computer-generated, rather than from a human (Lipnevich and Smith 2008).

In light of the challenges with feedback, as well as the somewhat limited and contradictory studies emerging around GenAI-generated feedback, it is valuable to understand at scale—across discipline and across university—how university students perceive and interact with AI-generated feedback. Furthermore, it is useful to compare students' perceptions of Generative AI feedback with their perceptions of comments provided by their teachers. For instance, do students treat AI and teacher-generated feedback with similar caution or criticality? How are they emotionally impacted, and how do they navigate this? Such comparisons may lead to understanding how, when and why AI and/or teacher-generated feedback could be used to support student learning.

Method

This paper reports on a subsection of a large-scale, cross-sectional survey that explored university students' use and perceived usefulness of AI for learning. In particular, this paper focuses on the survey items that probe for students' perceptions of AI-generated feedback compared to feedback from their educators. Specifically, the survey asks students if they have used GenAI for feedback and their perceptions of the helpfulness and trustworthiness of feedback from both AI and teachers. Additionally, open-ended questions ask students to describe the differences between the feedback received from AI and their teachers, including any differences in emotional impact and effects on learning.

Participants

Participation in this study was voluntary and anonymous. Approximately 192,000 students from four Australian universities (Deakin University, Monash University, University of Queensland, University of Technology Sydney) were invited to participate in the online survey, and 5% ($n = 10,132$) of students volunteered. From those, 79% ($n = 8028$) completed the survey. This paper specifically draws on data from 6960 students who responded to the survey item regarding using GenAI for feedback purposes. In brief, 57% were women, 72% were aged between 18 and 24 years, and 58% spoke English as the main language at home. The sample comprised 89% full time, 58% domestic, 61% undergraduate, and 92% on campus. 36% of students were from Sciences, Technology, Engineering and Mathematics, 24% from Health, 20% from Humanities and Social Sciences, and 20% from Business and Law.

Survey

The survey is published under a Creative Commons BY-NC-SA 4.0 license (Chung et al. 2024). The survey items were developed through multiple rounds of consultation with the literature and experts. In addition, the survey underwent a process of piloting and refinement to ensure content validity and usability. Staff and students from the participating universities tested the survey,

providing feedback on question clarity, comprehension, and usability (see Chung et al. 2024, for further details about the survey construction).

In this paper, we explore the nine questions specific to feedback (Table 1). All of the students who responded to these questions had already indicated that they had previously used GenAI for their university studies in some way. Before answering the questions about feedback, the students were provided the following description: ‘Feedback is when you get information that you are able to use to help you improve your work. This includes getting: suggestions for improving your work; details of the strengths and weaknesses of your work; suggested text edits; and additional ideas.’ This description is consistent with feedback as a student-centred process but framed in a way for students to quickly understand.

Data collection

Approval was received from the human research ethics committees from all four universities. Recruitment of students occurred in different ways across the four universities. In two of the universities, blanket emails were sent to students, at one university a pop-up was used within the learning management system, and the final university invited unit/subject coordinators to post the invitation to their students in their online learning management system. An incentive was offered to students for participation: the chance to win a \$350 gift card voucher (one prize per university). The survey was available between August and October 2024.

Quantitative analysis

Data was cleaned, processed and analysed using SPSS Version 29. The four feedback survey items were rated on a 5-point scale and treated as continuous data. Two paired samples *t*-tests were conducted to compare the means of the two helpfulness items (between GenAI and teacher) and similarly, the two trustworthiness items. This determined any statistically significant difference between pairs of ratings in helpfulness (and trustworthiness) of GenAI and teacher feedback.

Qualitative analysis

We conducted a thematic analysis of the open-ended responses, following the process described by Braun and Clarke (2006). An inductive approach was used, acknowledging that as feedback

Table 1. Survey questions.

Item	Question	Type
Q37	Have you used GenAI in your studies to get feedback on your work, such as assignments or Projects?	Yes/No <i>If students responded 'No' they would then jump to Q43</i>
Q38.1	How helpful was the feedback from GenAI	5-point likert scale: 1 = <i>very unhelpful</i> , 5 = <i>very helpful</i>
Q38.2	How helpful was the feedback from your teacher	
Q39.1	How trustworthy was the feedback from GenAI	5-point likert scale: 1 = <i>very untrustworthy</i> , 5 = <i>very trustworthy</i>
Q39.2	How trustworthy was the feedback from your teacher	
Q40	Please describe any differences in how it impacted your learning (comparing GenAI and your teacher)?	Open-ended text response
Q41	Were there any differences in how the feedback made you feel (comparing GenAI and your teacher)?	
Q42	Describe any other differences between the feedback you received from GenAI and your teacher.	
Q43	Why didn't you use GenAI for feedback on your work?	Open-ended text response <i>Only asked if students responded 'No' to Q37.</i>

researchers, we bring domain theory to the topic and actively construct themes (Varpio et al. 2017). Semantic coding (Braun and Clarke 2006) was utilised in order to describe the explicit or surface meanings of the data. Such an approach was considered to be most appropriate given the broad nature of the comments and the recognised complications of coding latent meanings from short open-ended survey responses (LaDonna, Taylor, and Lingard 2018).

Our coding framework was iteratively developed by three of the authors (MH, MB, JH) through a grounded, constant comparative method: the authors read subsets of the data, independently developed codes, and then negotiated an initial coding structure of 18 codes. The team, and a research assistant (XL) then used these codes to analyse a larger sample of the comments and negotiate new codes, using a constant comparative method until theoretical saturation was achieved. The coders (JH, XL) then continued with the coding, consulting with each other as a process of forming consensus and sustaining inter-coder agreement as described by Campbell et al. (2013). Inter-coder agreement was assessed subjectively similar to the process outlined by Guest, MacQueen, and Namey (2012). In addition, an audit trail (Guest, MacQueen, and Namey 2012) was kept by both coders, which helped the lead author (MH) to engage in internal replication (Dey 1993) of a random sample of codes from both coders, with any discrepancies leading to a further round of negotiation. In total, the coders met a further six times to compare coding and difficult responses. If a new code was proposed, this came back to the full team to discuss and review the data before being added to the code book. Through this process the initial 18 codes were refined and expanded, resulting in 48 codes organised in themes and subthemes. The first three open-ended questions (Table 1: Q40–42) were a series of prompts to help students elaborate on their comparative experience of GenAI and teacher feedback. As a result, 3 codes were allocated to identifying the subject of the comment (AI, Teacher, or both), and three codes allocated to identifying the comparator (more, less, similar) and 26 codes were used to identify the characteristic of the comparison (e.g. in-depth). In this way, each coded instance had a set of three codes: subject/actor (e.g. AI), comparator (e.g. more), characteristic (e.g. accessible). The analysis of the fourth open-ended question - asked of students who did not use GenAI for feedback - resulted in 16 codes. The full code book with definitions and example quotes is published here: <https://doi.org/10.26180/28861781>.

Results

Quantitative results show that student usage of GenAI for feedback was nearly evenly split, with 49.7% ($n=3461$) reporting they sought feedback from GenAI, while 50.3% ($n=3499$) did not. Students who used GenAI for feedback were then asked to compare its feedback to that of their teachers. 96% of the students ($n=3328$) answered these questions. A large majority found both sources somewhat or very helpful, with 83.9% saying this about AI feedback and 82.2% about teacher feedback. However, perceptions of trustworthiness were notably different—90.5% rated teacher feedback as somewhat or very trustworthy, compared to just 60.1% for AI feedback. Table 2 presents the means, standard deviations, and frequency ratings of helpfulness and trustworthiness. Figure 1 visualises the frequencies presented in Table 2, highlighting that although most students responded positively to both GenAI and teacher feedback, the latter elicited a notably stronger positive response, especially with regards to trustworthiness.

A paired samples *t*-test revealed a statistically significant difference between how helpful students rated GenAI feedback and teacher feedback, with a mean difference of 0.14 ($SD=1.15$), $t(3327) = -7.17$, $p < .001$, with a small effect size (Cohen's $d=-0.13$). Similarly, the second paired samples *t*-test revealed a statistically significant difference between how trustworthy students rated GenAI feedback and teacher feedback, with a mean difference of 0.93 ($SD=1.09$), $t(3327) = -49.30$, $p < .001$, with a large effect size (Cohen's $d=-0.89$). Students reported teachers providing more helpful feedback than GenAI, and similarly rated them as more trustworthy than GenAI.

Table 2. Helpfulness and trustworthiness of GenAI and teacher feedback.

	<i>M</i>	<i>SD</i>	How helpful was the feedback from				How trustworthy was the feedback from			
			Very unhelpful	Somewhat unhelpful	Neither helpful nor unhelpful	Somewhat helpful	Very helpful	Very untrustworthy	Somewhat untrustworthy	Neither trustworthy nor untrustworthy
How helpful was the feedback from	GenAI	4.02	0.79	41 (1.2%)	141 (4.2%)	352 (10.6%)	1981 (59.5%)	813 (24.4%)		
	Teacher	4.16	0.90	42 (1.3%)	165 (5%)	386 (11.6%)	1362 (40.9%)	1373 (41.3%)		
How trustworthy was the feedback from	GenAI	3.53	0.89	66 (2%)	405 (12.2%)	856 (25.7%)	1708 (51.3%)	293 (8.8%)		
	Teacher	4.46	0.74	19 (0.6%)	49 (1.5%)	248 (7.5%)	1071 (32.2%)	1941 (58.3%)		

N. 3328.

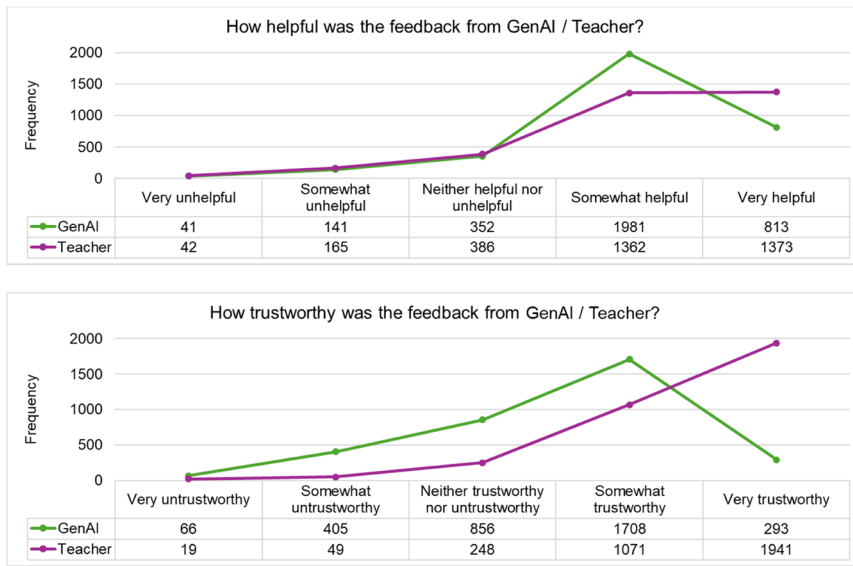


Figure 1. Visualising the response frequencies in Table 2, concerning the helpfulness and trustworthiness of feedback from GenAI and teachers.

Table 3. Qualitative coding totals.

	Used GenAI for feedback		Total
	Yes	No	
Students	3461	3499	6960
% of total students	49.7%	50.3%	100.0%
Students who responded to at least one open-ended question	2433 (70.3%)	2992 (85.5%)	5425 (77.9%)
Comments that could be coded	5736	2906	8642
Coded instances	8498	3405	11,903

Thematic analysis of open-ended responses

These quantitative descriptions of use, perceived helpfulness and perceived trustworthiness frame the qualitative findings, which provide a richer explanation of students' values and perceptions of GenAI and teacher feedback. Overall, 77.9% of the students provided at least one response to the open-ended questions, with a total of 8642 usable text responses. This resulted in 11,903 coded instances. Table 3 provides an overview of qualitative coding.

Comparing GenAI to teacher-provided feedback

Table 4 reports on the 6 themes, 5 subthemes and 26 characteristics which students described when comparing GenAI and Teacher provided feedback. While the original coding identified three agents (GenAI, Teacher, Both), this table reports on a combination of the data so that all statements are presented with GenAI being compared with the Teacher.

Out of the six themes the most frequently reported related to the nature of the feedback information (54.8%), and in particular the quality of that information. GenAI, in comparison to teacher-provided feedback, was considered less reliable, relevant and contextualised, but it was seen to be more objective (i.e. less biased), and offered more understandable information. The second most commonly reported theme was that related to feedback processes (22.7%). Within this

Table 4. Comparison of GenAI to teacher feedback.

			GenAI (in comparison with Teacher provided feedback)			Total	
			Less	Similar	More		
Theme & Subtheme	Characteristic (code) and short description	n	# (% row)	# (% row)	# (% row)	Sub Theme (% theme)	Theme (% column)
Processes							
Access	Ease (easily accessible)	875	6 (0.7%)		869 (99.3%)	1793 (93%)	1928 (22.7%)
	Speed (immediacy, fast)	614	8 (1.3%)		606 (98.7%)		
	Volume (can be asked a lot of questions)	304	7 (2.3%)		297 (97.7%)		
Timing	Before submission (when the interaction occurs)	31			31 (100%)	76 (3.9%)	
	After submission (when the interaction occurs)	45	45 (100%)				
Effort	Less effort (reduces effort, does my work)	59			59 (100%)	59 (3.1%)	
Sense-making							
	Understanding (helps me understand)	313	64 (20.4%)		249 (79.6%)	419 (100%)	419 (4.9%)
	Reflection (helps me to reflect on my work)	86	61 (70.9%)		25 (29.1%)		
	Progress (helps me to know how I am going)	20	10 (50%)		10 (50%)		
Information							
Quality	Specificity (detailed v. generic)	1118	544 (48.7%)	3 (0.3%)	571 (51.1%)	4392 (94.2%)	4660 (54.8%)
	In-depth (nuanced v. superficial)	254	158 (62.2%)		96 (37.8%)		
	Understandable (information that is easier to understand)	297	77 (25.9%)	1 (0.3%)	219 (73.7%)		
	Relevance (degree of ir/relevance of information)	123	104 (84.6%)		19 (15.4%)		
	Contextualised (sensitive to the specific context of the assignment and its requirements)	669	637 (95.2%)		32 (4.8%)		
	Utility (usable or helpful information)	590	250 (42.4%)	81 (13.7%)	259 (43.9%)		
	Reliable (accurate and trustworthy)	1143	1090 (95.4%)	3 (0.3%)	50 (4.4%)		
	Objective (objective v. subjective or biased)	198	15 (7.6%)	1 (0.5%)	182 (91.9%)		
	Tone	Positivity (positive, polite, agreeable)	188	24 (12.8%)	2 (1.1%)	162 (86.2%)	268 (5.8%)
Negativity (negative, dismissive, insulting)		80	67 (83.8%)	3 (3.8%)	10 (12.5%)		

(Continued)

Table 4. Continued.

		GenAI (in comparison with Teacher provided feedback)				Total	
			Less	Similar	More		
Feeling							
	Positive (makes me feel positive)	247	132 (53.4%)	10 (4%)	105 (42.5%)	471 (100%)	471 (5.5%)
	Negative (makes me feel negative)	109	93 (85.3%)	4 (3.7%)	12 (11%)		
	No Impact (has no impact on my feelings)	115		10 (8.7%)	105 (91.3%)		
Relational							
	Personal (connected, embodied, reciprocal)	471	442 (93.8%)		29 (6.2%)	896 (100%)	896 (10.5%)
	Risky (feels risky, vulnerable)	306	304 (99.3%)		2 (0.7%)		
	Expert (authority, expertise)	119	105 (88.2%)		14 (11.8%)		
Values							
	Importance (is important or valuable in some way)	124	111 (89.5%)	6 (4.8%)	7 (5.6%)	124 (100%)	124 (1.5%)
Total		8498	4354 (51.2%)	124 (1.5%)	4020 (47.3%)	8498	8498

theme there was a very strong indication that students felt GenAI provided better access, in terms of ability to get feedback anytime, quickly and in large volumes. Although the themes of sense-making, feeling, relational and values are not as strongly represented, the comments do reveal that while students perceive GenAI to be less personal (i.e. less relationally significant due to a lack of interpersonal connection) and less expert (i.e. less of an authority), at the same time GenAI is seen to be less risky (i.e. less likely to make them feel at risk or vulnerable), and less negative in tone and feeling.

Exploring why students chose not to use GenAI for feedback

Students who indicated that they had not used GenAI for feedback were asked a single open-ended question (Table 1: Q43). It should be noted that all of these students had previously indicated that they had used GenAI in some way for their university studies, but that they did not use it for feedback purposes. The 2906 comments revealed 16 reasons why they did not use GenAI for feedback purposes. While half of these appear similar to the characteristics identified by students who had used GenAI for feedback, the relationship is not necessarily the same. For example, 59 students who used GenAI for feedback purposes valued it because it reduced the effort required to do a task (e.g. it did it for them). In contrast, 18 students who did not use GenAI for feedback indicated that they did not use it because it could reduce the effort that they felt was an important part of the learning process (Table 5).

The most frequently noted reasons for not using GenAI for feedback related to concerns about the quality of the information (37.5%), and, in particular, its trustworthiness (28.7%). Interestingly, concerns with quality, and in particular, reliability, were also the most frequently noted for those students who did use GenAI for feedback. Another strong theme for those not using GenAI for feedback was that of feedback processes (29.3%). In particular, there was a large number of students ($n=957$, 28.1%) who indicated that they did not use GenAI for feedback because they did not know it was possible, or did not know how they might be able to do this.

Table 5. Reasons for not using GenAI for feedback.

Theme and Subtheme	Reason (code)	Description	Count (% column)	Theme (% column)
Processes				
Access Effort	Unaware	Unaware of the option or did not know how to.	957 (28.1%)	999 (29.3%)
	Effortful	Too much effort or not enough time to learn how to use GenAI.	24 (0.7%)	
	Less effort	Does not want to off-load what they see as part of the learning process.	18 (0.5%)	
Information				
Quality	Specificity	Lack of specific details.	4 (0.1%)	1280 (37.6%)
	In-Depth	Not enough, or sometimes not useful because it offers too much depth or additional explanation than desired.	32 (0.9%)	
	Contextualised	Does not know the context of the work, the feedback is not contextualised enough.	225 (6.6%)	
	Utility	Not helpful or useful.	40 (1.2%)	
Tone	Trustworthy	Not reliable or can't be trusted. Includes comments about trusting teachers, self or others more.	976 (28.7%)	3 (0.1%)
	Positivity	GenAI output is seen to be positive but in such a way that it raises doubt about its veracity, i.e. it is a people pleaser.		
	Relational			
	Personal	Values the role of a teacher, or connection with humans, preference for working with peers, etc.	204 (6%)	359 (10.5%)
	Expert	GenAI is not expert enough.	155 (4.6%)	
Values				
	Preference	General comments about preference to not use AI, or preference for other sources.	145 (4.3%)	767 (22.5%)
	Need	Either does not want feedback in the first place so there is no need, or does not see the need to use AI for feedback because they get it from self or others.	214 (6.3%)	
	Unsustainable	Impact on the environment in some way.	5 (0.1%)	
	Privacy	Concerns about data and privacy.	93 (2.7%)	
	Integrity	Concerns about breaking university rules, or a general feeling that it is not appropriate to use.	310 (9.1%)	
Total			3405	

It is also interesting to note that students who chose not to use GenAI for feedback made more comments than the other students around issues relating to values (22.5%), such as a general preference for human connection (4.3%) and a feeling there was no need to get GenAI feedback (6.3%) because they either did not want feedback, or they were satisfied with their current processes. A smaller number of students, but still noteworthy, also indicated concerns about privacy ($n=93$, 2.7%) and integrity ($n=310$, 9.1%).

Discussion

Students who used GenAI feedback

Half of our student participants reported using GenAI for feedback. This is promising because it reveals a large proportion of students actively engaging in feedback seeking behaviours which is regularly communicated in feedback literature as a valuable condition that supports effective feedback (Henderson, Molloy, et al., 2019; Winstone et al. 2017). Our findings highlight that GenAI feedback is already a common student-initiated practice, and as such there is a strong need for higher education institutions to consider how they might support the students to most effectively engage in this practice.

With half of the student body using GenAI for feedback, there is an urgent need to understand how useful they find it, why they find it *helpful*, and if they engage with it in critical ways, such as judging its *trustworthiness*. In terms of *helpfulness*, there is a statistically significant difference (small effect size), in which teachers are rated more helpful. However, this does not mean that students had a negative appraisal of GenAI's role in feedback. An overwhelming majority of the sample reported that feedback from both GenAI and teachers was helpful. When looking closer the strength of their perceptions is noteworthy: 24.4% of students rated GenAI feedback as *very helpful*, compared with 41.3% in relation to teacher feedback. We acknowledge that these are self-reported perceptions of helpfulness, however, our findings indicate that GenAI feedback, while helpful, was not seen to be as helpful as teacher feedback. This finding alone supports the notion that GenAI feedback should not be seen as a replacement for teacher feedback.

The nature of the helpfulness of these two sources of feedback is not necessarily the same – our data suggests they are valued for different reasons and are likely to serve different purposes. Overall, students indicated that both GenAI and Teacher feedback had *utility* and *specificity*. However, there was a strong indication that GenAI feedback was easier to *access* (*ease, speed, volume*), aided *sensemaking* insofar as it promoted *understanding*, and was felt to have some advantages in relation to some aspects of quality of information (*understandable and objective*). These findings affirm and further expand on the literature that claims GenAI can be a valuable feedback source (e.g. Chan and Hu 2023; Lee and Moore 2024; Shi and Aryadoust 2024). This finding of students regarding GenAI feedback as more *objective* compared with teacher feedback is in contrast with Nazaretsky et al. (2024) who found that students had lower evaluations of objectivity of AI-generated feedback. They conclude that students have a negative bias against GenAI as source of feedback information, suggesting a fundamental human preference for personal interaction. Our own data problematizes this broad claim since GenAI feedback is clearly valued by students for a number of reasons, of which objectivity is only a part.

While GenAI feedback is valued by students in some ways, teacher feedback was also reported to be more *relevant*, and *contextualised*. Similar results were reported by Ziqi et al. (2024), and appear related to an effect noted in Banihashem et al. (2024) study in which they compared GenAI and peer feedback. They noted that GenAI produced more extensively descriptive feedback including summary statements, whereas peers were better at identifying the specific issues that needed improving. This highlights the idea, arguably supported by our data, that teachers are likely to hold deeply contextual knowledge of the assessment, course and learner, and as a result are able to identify 'what matters'. Furthermore, we found that students valued teacher feedback as being more *personal*, and *expert*. This may be connected with Nazaretsky et al. (2024) findings in which humans were rated as more genuine, and more credible than AI.

These relational, contextually-expert and authoritative factors may underlie the striking difference in how students judged the *trustworthiness* of GenAI feedback to be less than that of teacher feedback (statistically significant, with large effect size). However, the experience of teacher feedback is not always positive. An interesting finding in our study was that GenAI feedback was felt to be *less risky* (in an interpersonal sense) when compared with teacher feedback. This aligns with Wang's (2024) study of language learners which found their anxiety around feedback was lower when feedback is from GenAI rather than a teacher. Our study also found that GenAI feedback was perceived to be more *positive* in tone (although this positivity was not always felt to be a good thing, with some students suggesting it felt disingenuous). On the other hand, teacher feedback was more likely to be noted as having a *negative tone* (e.g. dismissive, insulting) and result in *negative feelings*. Ajjawi et al. (2022) suggest that the negative emotions associated with feelings of incompetence, can be mediated by relational engagement. This suggests that teachers do not need to say positive things, but the tenor of their communication matters. This is noteworthy because negative experiences with feedback can have a large impact on further engagement (Ryan and Henderson 2018).

These comparisons of GenAI and teacher feedback again draw us to the idea that GenAI and teacher feedback may both be valued because they meet different needs. This directly challenges the claims that GenAI feedback is an opportunity for ‘liberating teachers from time-consuming tasks and allowing them to concentrate on higher-level responsibilities’ (Chan and Tsi 2024, p.2). While we think arguments for replacement are not well founded, we do think that GenAI has a potentially valuable role as an additional source of feedback.

Building on this, the literature already provides some insights into how GenAI and teacher feedback may be complementary rather than competing. For example, previous work has identified useful differences between the two, including the specificity of peer feedback in comparison to the comprehensive nature of GenAI feedback (Banihashem et al. 2024), the ability for human feedback to provide mutual recognition of each other as humans in a social world (Corbin, Tai, and Flenady 2025), a higher student agency when working with GenAI feedback (Jensen, Bearman, Boud, Konradsen 2025) and the more developmental focus of teacher feedback (Jensen et al. 2025). As outlined above, our findings add to this: noting that GenAI feedback is more *accessible*, more *understandable* and may invite engagement since it is perceived to be *less risky*. In contrast, teacher feedback may be more credible and insightful given that it is reported to be more *relational, contextualised, relevant and reliable*. Taking the literature and our findings together, we infer the following complementary relationship. At best, teacher feedback challenges students with contextualised areas for improvement that operate beyond the immediate task, with all the mutuality and recognition of each others’ lifeworlds that productive student-teachers relationships entail. This can be complemented by GenAI feedback, which presents helpful, digestible information about present student work, which is easy to understand and is almost always accessible, and without any risk of loss of social status.

We note a much smaller proportion regarded GenAI feedback as *very trustworthy*. There is a big proviso however, students themselves need to be able to understand if the GenAI’s comments on their work are in fact *trustworthy* or *reliable*. This cannot be based solely on an uninformed impression. Indeed, we should not assume all GenAI use is equal. If a student is in an environment rich with social and material interactions, as Esterhazy (2018) describes in her ethnography of student feedback interactions, then GenAI will be just one part of many feedback experiences. However, feedback rich environments are not common (Jensen et al. 2025) and as a result students may be relying heavily on GenAI output. Therefore, we agree with Bearman et al. (2024) that there is an urgent need to develop students’ evaluative judgement – or capability to judge the work of self and others – particularly around GenAI. This goes hand-in-glove with other aspects of feedback literacy, including understanding the need for multiple perspectives about a piece of work (Carless and Boud 2018).

Students who did not use GenAI feedback

Half (50.3%) of the students reported that they did not use GenAI for feedback. The analysis of their reasons provides further insight into student perspectives and key considerations for policy, support, and formal integration of GenAI feedback in higher education.

However, it should be noted that while these students did not use GenAI for feedback, they had all used GenAI in some way for their university studies. One reason, reported by 28.1%, was that they were *unaware* of the option to use GenAI for feedback purposes, or indicated that they did not know how to do so (e.g. how to upload or input a whole essay for feedback). This is concerning since 83.9% of students who did use GenAI for feedback reported it helpful or very helpful. We do not argue that GenAI feedback must be used. However, our data indicates that many students found it to be useful, and therefore all students should be supported in developing an understanding of how they might access and use it for this purpose.

Another commonly reported reason for not using GenAI was to do with *trustworthiness* of the feedback (28.7%), in particular, the outputs or processes of GenAI were not regarded as

reliable, or not as trustworthy as teachers, self or peers. Interestingly, a similar number of students who used GenAI feedback were also concerned about its *reliability* (31.5%). This echoes the reports of students who are using GenAI for feedback. Across both groups, some students cite it as a reason for not using GenAI, while others merely note it as a caution. Upon reflection, we wonder if these percentages are too small. Should more students be critical of GenAI? Arguably, students should bring to bear their critical judgment of the quality and appropriateness of any feedback regardless of its source. On the other hand, it is unclear exactly how the students have come to form these evaluations. Perhaps they are less related to the efficacy of the system or other issues of quality, but more reflective of a general attitude towards credibility of source (Lipnevich and Smith 2008; Nazaretsky et al. 2024). On the other hand, perhaps the perceptions of trustworthiness are reflecting a judicious discernment in relation to the reliability of the feedback across all contexts. In other words, perhaps our data reflects students' recognition that GenAI can be trusted in some instances more than others, such as the difference between asking GenAI to provide advice about paragraph structures compared with seeking guidance that requires a deeply contextualised understanding of the course content. If this is the case, then the obvious conclusion, as we have noted previously, is that GenAI feedback and teacher feedback are useful for different reasons, that is, they are complementary but are not interchangeable. Clearly there is a need to continue to research students' perception of trust with respect to feedback as engagement with GenAI continues to develop.

Future work

The insights from students who used and those who didn't use GenAI for feedback point to future work on feedback literacy and feedback engagement. Zhan and Yan (2025) have investigated feedback literacy in the context of English language testing. They identify that student feedback literacy in a time of GenAI includes: the capability to write and finetune feedback prompts; evaluative judgement about GenAI outputs; ethical decision-making in terms of how to use the GenAI feedback; emotional reflexivity to calibrate trust levels; and employing metacognitive skills in terms of regulating their own feedback processes with respect to GenAI. From a conceptual perspective, Zhan et al. (2025) propose that feedback engagement is a means of reconciling these feedback skills with their environment in terms of eliciting, processing and enacting feedback. This framing is helpful, as although our work points to why students seek or elicit feedback from genAI – it is helpful, understandable, less risky, approachable and they were aware of its capabilities – we have less of a sense of how they process and enact feedback into their future work. What happens when they try to integrate what they have learnt through GenAI feedback into future tasks or modes of thinking? Do GenAI and teacher feedback continue to complement each other? GenAI feedback tends to be available before an assessment is due rather than after – what difference does this make? Does the bias against automated feedback (Nazaretsky et al. 2024) mean students process and enact this feedback differently? These all form areas for future work.

Limitations

This is a large-scale study drawn from four Australian institutions. However, we note that the response rate of 8,000 given the broad enrolment numbers of over 200,000 students cannot claim to be representative. The survey was optional and therefore students with stronger views about GenAI may have been more inclined to answer the survey. We also note that Australian higher education may differ from other contexts; our demographics however give a sense of the key features of the respondents. Finally, this survey was completed at a point where very few

students would have experienced GenAI feedback explicitly grounded in their curriculum/assignment resources, a development which will likely change as universities provide secure, customised GenAI. This could change students' perceptions significantly.

Conclusion

This study underlines that GenAI and teacher feedback are perceived and experienced differently by university students in their day-to-day study. Students regard teacher feedback as relatively more helpful and more trustworthy. But their comments suggest that GenAI feedback is not unhelpful nor altogether untrustworthy. GenAI feedback is considerably easier to access, is more understandable, less negative and aids sense-making (understanding). And while engaging with GenAI for feedback provides a lesser sense of connection, it also affords a lessened personal risk. Overall, the findings suggest that these feedback modes are not an either/or; students can and do engage with both, and there are likely benefits from doing so because they can helpfully serve different functions. The ways in which students perceive GenAI and teacher feedback provide important insights into not only how we enact GenAI feedback in institutions but also how we might further understand the value, usefulness, and relational impact of teacher feedback practice. The challenge for the sector is to explore how teachers can lean into what they do best, contextualised, expert and challenging advice, in the context of the teacher-learner relationship.

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The authors report there are no competing interests to declare.

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